

```
Sep 15 18:44
saad@sand-dweller: ~/xv6-labs-2026 — make qemu
~/xv6-labs-2026

? How would you like to authenticate GitHub CLI? Login with a web browser.

! First copy your one-time code: A091-3734
Press Enter to open github.com in your browser...
Opening in existing browser session.
✓ Authentication complete.
- gh config set -h github.com git_protocol https
✓ Configured git protocol
✓ Logged in as dumb-monke
saad@sand-dweller:~$ cd xv6-labs-2026/
saad@sand-dweller:~/xv6-labs-2026$ git push -u origin util
Enumerating objects: 8238, done.
Counting objects: 100% (8238/8238), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3516/3516), done.
Writing objects: 100% (8238/8238), 17.38 MiB | 2.38 MiB/s, done.
Total 8238 (delta 4697), reused 8238 (delta 4697), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4697/4697), done.
To https://github.com/dumb-monke/saad31625-xv6-labs.git
 * [new branch]      util -> util
branch 'util' set up to track 'origin/util'.
saad@sand-dweller:~/xv6-labs-2026$ git branch
* util
saad@sand-dweller:~/xv6-labs-2026$ git switche problemSet1
git: 'switche' is not a git command. See 'git --help'.

The most similar command is
      switch
saad@sand-dweller:~/xv6-labs-2026$ git switch problemSet1
fatal: invalid reference: problemSet1
saad@sand-dweller:~/xv6-labs-2026$ git switch problemSet1
fatal: invalid reference: problemSet1
saad@sand-dweller:~/xv6-labs-2026$ git switch util
git switch -c problemSet1
Already on 'util'
Your branch is up to date with 'origin/util'.
Switched to a new branch 'problemSet1'
saad@sand-dweller:~/xv6-labs-2026$ git branch
* problemSet1
  util
saad@sand-dweller:~/xv6-labs-2026$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img READEME user/findtest.sh user/sixfive.txt user/cat user/echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombi
e user/_logstress user/_forphan user/_dorphan user/_sync user/_nendump
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first bitmap blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0
xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$
```

\$ yeah so sleep like shows no output to show it working , taking screenshot anyway


```

Sep 15 20:27
saad@sand-dweller: ~/xv6-labs-2026 — make qemu
hart 2 starting
hart 1 starting
init: starting sh
$ sleep 10
$ sixfive sixfive.txt
5
100
10
0
$ mendum
Example 1:
01010
2020
Example 2:
a string
Example 3:
another
Example 4:
020
1019430067
100
z
xyzzy
Example 5:
hello
w
o
r
l
d
$ $ echo deadcode | mendum hcccc
exec $ failed
mendum: not enough data for 'h'
$ mendum
Example 1:
01010
2020
Example 2:
a string
Example 3:
another
Example 4:
020
1019430067
100
z
xyzzy
Example 5:
hello
w
o
r
l
d
$
```

```
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_sync user/sync.o user/ulib.o user/usys.o user/print.o user/unalloc.o
riscv64-linux-gnu-objdump -t user/_sync | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/sync.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_UTIL -DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-com
mon -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-bu
iltin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/sleep.o user/sleep.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_sleep user/sleep.o user/ulib.o user/usys.o user/print.o user/unalloc.o
riscv64-linux-gnu-objdump -t user/_sleep > user/sleep.asm
riscv64-linux-gnu-objdump -t user/_sleep | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/sleep.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_UTIL -DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-com
mon -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-bu
iltin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/sixfive.o user/sixfive.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_sixfive user/sixfive.o user/ulib.o user/usys.o user/print.o user/unalloc.o
riscv64-linux-gnu-objdump -t user/_sixfive > user/sixfive.asm
riscv64-linux-gnu-objdump -t user/_sixfive | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/sixfive.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_UTIL -DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-com
mon -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-bu
iltin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/memdump.o user/memdump.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_memdump user/memdump.o user/ulib.o user/usys.o user/print.o user/unalloc.o
riscv64-linux-gnu-objdump -t user/_memdump > user/memdump.asm
riscv64-linux-gnu-objdump -t user/_memdump | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/memdump.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_UTIL -DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mmodel=medany -ffreestanding -fno-com
mon -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-bu
iltin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/find.o user/find.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_find user/find.o user/ulib.o user/usys.o user/print.o user/unalloc.o
riscv64-linux-gnu-objdump -t user/_find > user/find.asm
riscv64-linux-gnu-objdump -t user/_find | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/find.sym
mkfs/mkfs fs.img RD4ME user/finctest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombi
e user/_logstress user/_forphan user/_dorphan user/_sync user/_sleep user/_sixfive user/_memdump user/_find user/_memdump
meta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloot: first 1180 blocks have been allocated
balloot: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ find . b
$ echo > b
$ mkdir a
$ echo > a/b
$ mkdir a/aa
$ echo > a/aa/b
$ find . b
./b
./a/b
$ echo > a/aa/b
$ find . b
./b
./a/b
./a/aa/b
$
```

```
Sep 15 21:06
saad@sand-dweller: ~/xv6-labs-2026 — make qemu
~/xv6-labs-2026

riscv64-linux-gnu-objdump -S user/forphan > user/forphan.asm
riscv64-linux-gnu-gcc -t user/forphan | sed '1,/SYMBOL TABLE/d; s/ .* / /; /$$/d' > user/forphan.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026- -DSOL_UTIL -DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mcontrol-medany -ffreestanding -fno-com
mon -nostdlib -fno-builtin-strcmp -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-bu
lltin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/forphan.o user/forphan.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/forphan user/forphan.o user/ulib.o user/usys.o user/printf.o user/unalloc.o
riscv64-linux-gnu-objdump -S user/forphan > user/forphan.asm
riscv64-linux-gnu-gcc -t user/forphan | sed '1,/SYMBOL TABLE/d; s/ .* / /; /$$/d' > user/forphan.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026- -DSOL_UTIL -DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mcontrol-medany -ffreestanding -fno-com
mon -nostdlib -fno-builtin-strcmp -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-bu
lltin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/sync.o user/sync.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/sync user/sync.o user/ulib.o user/usys.o user/printf.o user/unalloc.o
riscv64-linux-gnu-objdump -S user/sync > user/sync.asm
riscv64-linux-gnu-objdump -t user/sync | sed '1,/SYMBOL TABLE/d; s/ .* / /; /$$/d' > user/sync.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026- -DSOL_UTIL -DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mcontrol-medany -ffreestanding -fno-com
mon -nostdlib -fno-builtin-strcmp -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-bu
lltin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/sleep.o user/sleep.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/sleep user/sleep.o user/ulib.o user/usys.o user/printf.o user/unalloc.o
riscv64-linux-gnu-objdump -S user/sleep > user/sleep.asm
riscv64-linux-gnu-objdump -t user/sleep | sed '1,/SYMBOL TABLE/d; s/ .* / /; /$$/d' > user/sleep.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026- -DSOL_UTIL -DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mcontrol-medany -ffreestanding -fno-com
mon -nostdlib -fno-builtin-strcmp -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-bu
lltin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/sixfive.o user/sixfive.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/sixfive user/sixfive.o user/ulib.o user/usys.o user/printf.o user/unalloc.o
riscv64-linux-gnu-objdump -S user/sixfive > user/sixfive.asm
riscv64-linux-gnu-objdump -t user/sixfive | sed '1,/SYMBOL TABLE/d; s/ .* / /; /$$/d' > user/sixfive.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026- -DSOL_UTIL -DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mcontrol-medany -ffreestanding -fno-com
mon -nostdlib -fno-builtin-strcmp -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-bu
lltin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/memdup.o user/memdup.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/memdup user/memdup.o user/ulib.o user/usys.o user/printf.o user/unalloc.o
riscv64-linux-gnu-objdump -S user/memdup > user/memdup.asm
riscv64-linux-gnu-objdump -t user/memdup | sed '1,/SYMBOL TABLE/d; s/ .* / /; /$$/d' > user/memdup.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026- -DSOL_UTIL -DLAB_UTIL -march=rv64gc -std=gnu99 -MD -mcontrol-medany -ffreestanding -fno-com
mon -nostdlib -fno-builtin-strcmp -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-bu
lltin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/find.o user/find.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/find user/find.o user/ulib.o user/usys.o user/printf.o user/unalloc.o
riscv64-linux-gnu-objdump -S user/find > user/find.asm
riscv64-linux-gnu-objdump -t user/find | sed '1,/SYMBOL TABLE/d; s/ .* / /; /$$/d' > user/find.sym
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/cat user/echo user/forktest user/grep user/init user/kill user/_ls user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombi
e user/_logstress user/_forphan user/_dorphan user/_sync user/_sleep user/_sixfive user/_memdup user/_find user/_memdup
mmap 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1192 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ sh < findtest.sh
$ $ $ $ $ hello
hello
hello
$ $
```